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The Editor
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Dear Editor,

We are pleased to submit our manuscript, *FloodLite: A CPU-and-Browser Edge-Deployment Benchmark for Lightweight Flood Segmentation*, for consideration as a Research Article in the *International Journal of Disaster Risk Reduction*.

Flood-segmentation models are almost always judged by accuracy on a benchmark. The responders who would use them run laptops, tablets and browsers, where latency, quantization behaviour and false alarms decide whether a model is usable. Apart from timings on NVIDIA Jetson boards, the flood-segmentation literature does not, to our knowledge, report laptop-CPU or in-browser latency together with quantization stability. Our manuscript measures them for three lightweight UNet students and a UNet+EfficientNet-B0 teacher, on an Apple Silicon CPU and in browser WebAssembly, in FP32 and INT8.

Three findings make the work relevant to disaster-risk practice:

1. Flood-only training data produces models that cannot say "no flood". The public Flood Semantic Segmentation Dataset (FSSD) contains no water-free images. Evaluated zero-shot on overlap-verified external data, every FSSD-trained FP32 model predicts water on 70–100% of dry scenes, a failure invisible to every FSSD metric and disqualifying for unattended monitoring. Retraining on a combined public corpus that includes water-free images cuts student false-alarm rates to 3–9%.

2. Distillation helps only with headroom, and it transmits the teacher's errors. On saturated FSSD, no response-, feature- or combined-distillation configuration improves on the no-distillation baseline. On the less saturated combined corpus, combined distillation raises flood-only IoU significantly for all three students, but distilled students also false-alarm on the teacher's false-alarm scenes more than twice as often as undistilled ones. Teachers should be audited for the deployment-relevant failure before they are used for distillation.

3. INT8 post-training quantization is architecture-dependent. Under one static-QDQ recipe, EfficientNet-Lite0 loses 0.043 IoU while MobileNetV3-Small (HardSwish) and MobileViT-XXS (attention) collapse by 0.45–0.46 IoU, and quantized models degrade further under domain shift. The recommended configuration, EfficientNet-Lite0-UNet INT8 on ONNX Runtime CPU, runs at 58 FPS on a 2023 laptop CPU, 2.2× faster and about 4× smaller than the teacher on the same runtime; in the browser, FP32 students reach 6–7.5 FPS (INT8 was not measured there).

The work fits IJDRR's scope because its contribution is deployment evidence for disaster-response practice rather than a new network design: the quantities we report are the ones a humanitarian

data-science team needs to decide whether, where and how a flood-segmentation model can be fielded, including the data it must be trained on to be trusted unattended.

Reproducibility and openness. All code, ONNX exports, per-fold checkpoints, and benchmarking harnesses are released under MIT licence at `https://github.com/NoorTasfia/flood-segmentation`. A Zenodo-archived snapshot with a DOI will be minted at submission. The FSSD dataset is publicly available on Kaggle.

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We would be grateful for your consideration of this submission and welcome any feedback from the editorial team and reviewers.

Sincerely,

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On behalf of all authors